

Amin Dehghani

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SUMMARY

Process Engineering Technician with hands-on experience in high-volume PCBA/SMT manufacturing, processing equipment setup, and line change configuration. IPC-A-610 and IPC J-STD-001 certified with proven ability to read and interpret electronic schematics, assembly drawings, and Gerber files for process setup and defect resolution. Experienced in monitoring production lines, auditing process parameters, and providing technical guidance to line operators. Skilled in processing test runs, preparing production records (breakdown events, scrap, yield data), and presenting SPC charts and trend analysis to engineering for yield improvement. Strong corrective maintenance background with demonstrated leadership in cross-functional manufacturing environments. Proficient with automated PCBA/SMT equipment including ERSA, Electrovert, Versaflow, and Yamaha systems.

CORE COMPETENCIES

Processing Equipment Setup	Line Change Configuration	PCBA / SMT Processes	Corrective Maintenance	Line Operator Guidance	
Schematic Reading	SPC & Yield Analysis	IPC-A-610 / J-STD-001	Automated Equipment	Escalation Procedures	Test Run Processing
Cross-Functional Leadership					

EXPERIENCE

SolarEdge - Austin, TX

2024 - Present

Process Technician / Electronics Manufacturing Technician

Supporting high-volume PCBA/SMT assembly operations on the SolarEdge production line.

- Performed set-up of processing equipment including ERSA and Electrovert wave and selective soldering systems, configuring solder parameters, flux application, and conveyor settings for each product run.
- Executed line change configurations for new product introductions, coordinating feeder setup, component placement verification, and process parameter adjustments on Yamaha SMT automated equipment.
- Monitored, audited, and provided guidance to line operators on IPC standards (IPC-A-610, IPC J-STD-001), soldering quality, and correct process procedures.
- Processed test runs and prepared detailed records of breakdown events, scrap rates, and defect data; generated SPC charts and trend graphs for presentation to engineering to identify yield problems and validate new products.
- Provided technical assistance to process development and engineering personnel, supporting defect isolation using schematic analysis, in-circuit test (ICT) results, and AOI/SPI/X-ray inspection data.
- Read and interpreted electronic schematics, assembly drawings, and Gerber files to guide equipment setup, troubleshooting, and root cause analysis for engineering review.
- Performed corrective maintenance on production equipment, including basic troubleshooting, preventive maintenance, and equipment calibration to minimize downtime and support continuous production flow.
- Followed escalation procedures for process deviations and equipment failures, documenting issues and escalating to manufacturing engineering when outside standard operating parameters.
- Operated automated inspection systems including Eagle Eye 3D AOI, Pemtron SPI, and Phoenix Live X-ray for solder defect identification, void detection, and component misalignment verification.
- Operated PCB router/depaneling machines and supported ICT machine setup, ensuring accurate electrical validation of assembled PCBAs.
- Contributed to CAPA activities and yield improvement initiatives through data collection, SPC trend analysis, and root cause analysis (RCA).

Vapayesh Sanaat Fars - Iran

2012 - 2016

Hardware Technician / Maintenance Team

- Performed corrective and preventive maintenance on electrical, mechanical, and electromechanical systems.
- Diagnosed failures in control panels, motor drives, and mechanical assemblies; followed escalation procedures for complex repairs.
- Implemented preventive maintenance schedules, reducing equipment downtime by ~30%.
- Documented inspections, breakdown events, and modifications for compliance and long-term reliability.

Bushehr Nuclear Power Plant - Iran	2017
Systems & Risk Engineering Adviser	
• Evaluated operational risks in safety-critical environments and documented mitigation strategies. • Conducted system integrity assessments and supported technical process documentation.	

Javanan Sharq - Iran (Multiple Roles)	2016 - 2023
Software & Systems Engineer / Team Lead	
• Led cross-functional teams from project ideation through deployment; mentored junior engineers. • Built monitoring and automation pipelines using Python, C#, .NET, Docker, and Kubernetes. • Managed backend systems for Iran's largest e-commerce marketplace (DigiKala, \$12B+ valuation). • Established CI/CD pipelines reducing deployment times by 50% (Raimo Studio).	

SKILLS & CERTIFICATIONS

PCBA/SMT & Process Engineering: High-Volume PCB Assembly - Wave & Selective Soldering - SMT Processes - Processing Equipment Setup & Configuration - Line Change Configuration - Automated Equipment Operation (ERSA, Electrovert, Versaflow, Yamaha) - Schematic Reading & PCB-Level Debugging - Hand Soldering & Rework - Process Qualification & New Line Bring-Up	
Quality & Data: SPC Charts & Trend Analysis - Test Run Processing & Record Preparation - Yield Optimization & Defect Reduction - Root Cause Analysis (RCA) & CAPA - Metrology & Inspection (Pemtron SPI, Eagle Eye 3D AOI, X-ray) - SOP Development	
Maintenance & Troubleshooting: Corrective & Preventive Maintenance - Electrical, Mechanical, Pneumatic System Troubleshooting - Equipment Calibration & Tool Health Monitoring - Escalation Procedures	
Leadership & Systems: Line Operator Monitoring & Guidance - Cross-Functional Team Leadership - Process Data Analysis - Automation & Monitoring Support - Python, C#, SQL	
Certifications: IPC-A-610 (Acceptability of Electronic Assemblies) - IPC J-STD-001 (Soldering) - Docker & Kubernetes (2024) - AWS & Azure Systems (2024)	

EDUCATION

Bachelor of Science in Computer Engineering (GPA 3.6)	2012 - 2016
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